



Yakeen Plus (2025)

SHORT PRACTICE TEST - 01

DURATION : 60 Minutes

DATE : 11/09/2024

M. MARKS : 192

ANSWER KEY

PHYSICS

1. (3)
2. (1)
3. (3)
4. (3)
5. (4)
6. (4)
7. (1)
8. (3)
9. (4)
10. (3)
11. (3)
12. (1)

CHEMISTRY

13. (1)
14. (2)
15. (3)
16. (1)
17. (3)
18. (1)
19. (3)
20. (2)
21. (1)
22. (2)
23. (4)
24. (2)

BOTANY

25. (1)
26. (2)
27. (4)
28. (1)
29. (3)
30. (4)
31. (4)
32. (3)
33. (2)
34. (2)
35. (3)
36. (2)

ZOOLOGY

37. (1)
38. (4)
39. (4)
40. (2)
41. (3)
42. (3)
43. (1)
44. (1)
45. (2)
46. (4)
47. (1)
48. (3)

SECTION-(I) PHYSICS

1. (3)

$$S = \frac{a}{1-r} = \frac{1}{1-\frac{1}{2}}$$

$$= a=1, r=\frac{1}{2}$$

2. (1)

$$-\sqrt{8^2 + (-6)^2} \leq 8\sin\theta - 6\cos\theta \leq \sqrt{8^2 + (-6)^2}$$

$$-10 \leq 8\sin\theta - 6\cos\theta \leq 10$$

Then greatest value = 10

3. (3)

 $P - Q$, no subtraction is there because of different dimension.

4. (3)

$$3x - 3y - 1 = 0$$

$$3y = 3x - 1$$

$$\boxed{y = x - \frac{1}{3}}$$

5. (4)

$$\log_3 x^2 = 4$$

$$x^2 = 3^4$$

$$x = 3^2 = 9$$

6. (4)

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 1 - 2\sin^2 A = 2\cos^2 A - 1$$

7. (1)

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

8. (3)

$$\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 = x + \frac{1}{x} + 2$$

$$\frac{d}{dx}\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 = 1 - \frac{1}{x^2}$$

9. (4)

$$|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$$

$$\Rightarrow |\vec{A} + \vec{B}|^2 = |\vec{A} - \vec{B}|^2$$

$$A^2 + B^2 + 2\vec{A} \cdot \vec{B} = A^2 + B^2 - 2\vec{A} \cdot \vec{B}$$

$$4\vec{A} \cdot \vec{B} = 0$$

$$AB \cos\theta = 0$$

$$\cos\theta = 0$$

$$\theta = 90^\circ$$

10. (3)

Ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

11. (3)

$$R_{\max} = A + B$$

$$\theta = 0^\circ \therefore R_{\max} = 12 + 8 = 20N$$

12. (1)

$$P = \frac{a - t^2}{bx}$$

$$[a] = [T^2]$$

$$[ML^{-1}T^{-2}] = \frac{[T^2]}{[b][L]}$$

$$\Rightarrow [b] = \frac{[T^2]}{[ML^{-1}T^{-2}][L]}$$

$$= [M^{-1}T^4]$$

$$\left[\frac{a}{b}\right] = \frac{[T^2]}{[M^{-1}T^4]} = [MT^{-2}]$$

SECTION-(II) CHEMISTRY

13. (1)



$$2 + 2x - 2 \times 7 = 0;$$

$$2x - 14 + 2 = 0$$

$$2x = 12; x = \frac{12}{2} = +6.$$

[New NCERT Class 11th Page No. 239]

14. (2)



$$\text{Empirical formula mass} = 12 + 2 + 16 = 30$$

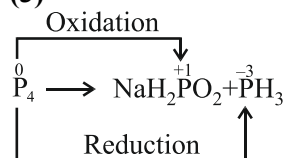
$$\text{Mol. Mass} = 2 \times \text{V.D.} = 2 \times 30 = 60$$

$$n = \frac{\text{Mol. mass}}{\text{empirical formula mass}} = \frac{60}{30} = 2$$

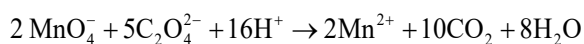
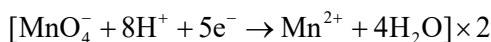
$$\text{Molecular formula} = (\text{Empirical formula})_n \\ = (CH_2O)_2 = C_2H_4O_2.$$

[New NCERT Class 11th Page No. 16]

15. (3)

[New NCERT Class 11th Page No. 244]

16. (1)



Thus, the coefficient of MnO_4^- , $\text{C}_2\text{O}_4^{2-}$ and H^+ in the above balanced equation respectively are 2, 5, 16.

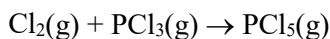
[New NCERT Class 11th Page No. 243]

17. (3)

- 1 g-atom of P = 31 g
- 2 mole of water = 36 g
- 22.4 L of CO_2 gas at STP = 44 g
- 6.02×10^{23} atoms of sulphur = 32 g

[New NCERT Class 11th Page No. 18]

18. (1)



10 mL of Cl_2 gas will produce 10 mL $\text{PCl}_5(\text{g})$

[New NCERT Class 11th Page No. 17]

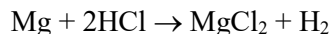
19. (3)

$$\text{Average atomic mass} = \frac{m_1x_1 + m_2x_2}{x_1 + x_2}$$

$$= \frac{35 \times 25 + 37 \times 75}{25 + 75} = 36.5$$

[New NCERT Class 11th Page No.13]

20. (2)



24 g Mg evolves 22.4 L H_2 at STP

$$12\text{ g Mg evolves } \text{H}_2 \text{ at STP} = \frac{22.4}{24} \times 12 = 11.2\text{ L}$$

[New NCERT Class 11th Page No. 14]

21. (1)

(i) Oxidation states of oxygen in O_2F_2 is +1

(ii) Oxidation states of oxygen in KO_2 is $-\frac{1}{2}$

[New NCERT Class 11th Page No. 246]

22. (2)

$$\text{Moles} = \text{Molarity} \times \text{Volume of solution in litres} \\ = 0.35 \times 0.25$$

$$\text{Mass} = \text{moles} \times \text{molar mass} \\ = 0.35 \times 0.25 \times 82 = 7.18\text{ g}$$

[New NCERT Class 11th Page No. 20]

23. (4)

In ClO_4^- , Cl is present in its highest oxidation state +7, so it can not show disproportion reaction.

[New NCERT Class 11th Page No. 241]

24. (2)

$$M = \frac{n_{(\text{NaOH})}}{V_{(\text{sol})}(\text{in L})} \Rightarrow 3 = \frac{\left(\frac{84}{40}\right)}{V_{(\text{sol})}(\text{in L})}$$

$$V_{(\text{sol})}(\text{in L}) = 0.7\text{ L}$$

[New NCERT Class 11th Page No. 20]

[Short Practice Test-01 | Yakeen Plus (2025) | 11/09/2024]

SECTION-(III) BOTANY

25. (1)

The cylindrical structure with a "cartwheel appearance" described here refers to the centriole. They are made up of nine evenly spaced peripheral fibrils of tubulin protein.

[NCERT Class 11th Page No. 99-100]

26. (2)

- The detailed structure of cell membrane was studied after the advent of electron microscopy.
- In human beings, the membrane of the erythrocyte has approximately 52 per cent protein and 40 per cent lipids.

[NCERT Class 11th Page No. 93]

27. (4)

The given figure represents the endoplasmic reticulum (ER), which includes the rough ER (RER) and the smooth ER (SER).

Golgi apparatus consist of many flat, disc-shaped sacs or cisternae of $0.5\mu\text{m}$ to $1.0\mu\text{m}$ diameter.

[NCERT Class 11th Page No. 95]

28. (1)

- The vacuole is bound by a single membrane called tonoplast.
- Ribosomes are composed of ribonucleic acid (RNA) and proteins and are not surrounded by any membrane.
- The content of nucleolus is continuous with the rest of the nucleoplasm.
- Prokaryotic flagella are structurally different from that of the eukaryotic flagella.

[NCERT Class 11th Page No. 96-102]

29. (3)
The number of mitochondria in a cell is higher in cells that are more physiologically active because those cells require more ATP for their energy needs, which is produced by the mitochondria
[NCERT Class 11th Page No. 97]

30. (4)
Chromatin stained by basic dyes.
[NCERT Class 11th Page No. 95,96,100]

31. (4)

- Sodium-potassium pumps move across cell membrane by active transport.
- Facilitated diffusion is a passive process that does not require energy, while active transport is an energy-requiring process.

[NCERT Class 11th Page No. 94]

32. (3)

Aleuroplasts	Protein
Leucoplasts	Colourless plastid
Amyloplasts	Potato
Elaioplasts	Oils and fats
Chromoplasts	Xanthophylls

[NCERT Class 11th Page No. 98-99]

33. (2)
The chloroplast length and width is 5-10 μm and 2 - 4 μm respectively.
Chlamydomonas contain 1 chloroplast per cell.
[NCERT Class 11th Page No. 97, 98]

34. (2)

Chloroplast	Contain chlorophyll and carotenoid pigments
Plasmodesmata	Connects cytoplasm of adjacent cells
Mitochondria	Sites of aerobic respiration
Cell wall	Provides barrier to undesirable macromolecules

[NCERT Class 11th Page No. 94, 97]

35. (3)
Rudolf Virchow modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape.
The bacterial cells, have a chemically complex cell envelope. The cell envelope consists of a tightly bound three layered structure i.e., the outermost glycocalyx followed by the cell wall and then the plasma membrane.
[NCERT Class 11th Page No. 89, 90]

36. (2)

Slime layer	Loose sheath
Capsule	Tough
Mesosomes	DNA replication
Pili	Conjugation
Fimbriae	Bristle like
Chromatophores	Contains pigments
Flagella	Motility

[NCERT Class 11th Page No. 90-91]

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SECTION-(IV) ZOOLOGY

37. (1)

The structures used for storing of food.	Crop
Ring of 6-8 blind tubules at junction of foregut and midgut.	Gastric caeca
Ring off 100-150 yellow coloured thin filaments at junction of midgut and hindgut.	Malpighian tubules
The structures used for grinding the food.	Gizzard

[Old NCERT Class 11th Page No. 113]

38. (4)

- Ligaments are a type of dense regular connective tissue. They connect bones to other bones, providing stability and support.
- Cartilage is a type of specialised connective tissue that is not classified as dense. It is

composed of cells called chondrocytes which are embedded in a matrix of collagen and elastic fibers.

[Old NCERT Class 11th Page No. 103]

39. (4)

- Neuroglia are the supportive cells of nervous tissue.
- Bone and blood are specialised type of connective tissues.
- Adipose tissue is a type of loose connective tissue.

[Old NCERT Class 11th Page No. 103]

40. (2)

- Tegmina or forewings (the first pair of wings) in cockroach arises from mesothorax.
- No wing arises from prothorax and hindwings arise from metathorax.

[Old NCERT Class 11th Page No. 112]

41. (3)

Malpighian tubules of cockroaches are used for excretion.

[Old NCERT Class 11th Page No. 113]

42. (3)

The correct route for transport of sperms in male frog is:

Testes → Vasa efferentia → Kidney → Bidder's canal → Urinogenital duct → Cloaca.

[Old NCERT Class 11th Page No. 119]

43. (1)

Gap junctions facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules.

[Old NCERT Class 11th Page No. 102]

44. (1)

In frogs, the hindlimbs end in five digits and they are larger and muscular than forelimbs that end in four digits.

[Old NCERT Class 11th Page No. 116]

45. (2)

- Bones are the hardest tissue of our body that support and protect softer tissues and organs.
- Adipose tissue is a type of loose connective tissue specialised to store fats.
- Cuboidal epithelium is found in ducts of glands and tubular parts of nephrons in kidneys.
- Neuroglia make up more than one half the volume of neural tissues in our body.
- Intercalated disc is the characteristic feature of cardiac muscle tissues.

[Old NCERT Class 11th Page No. 103, 104, 105]

46. (4)

In frogs, cloaca is used to pass faecal matter, gametes and urine to the exterior.

[Old NCERT Class 11th Page No. 119]

47. (1)

- Compound epithelium is made of more than one layer (multi-layered) of cells and thus has a limited role in secretion and absorption.
- Their main function is to provide protection against chemical and mechanical stresses. They cover the dry surface of the skin, the moist surface of buccal cavity, pharynx.

[Old NCERT Class 11th Page No. 102]

48. (3)

Columnar epithelium are found in the lining of stomach and intestine and help in secretion and absorption.

[Old NCERT Class 11th Page No. 101]

